

535514 Reinforcement Learning

Pre-Lecture Assignment: Lecture 4

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Problem: Policy Gradient Implemented in Python

In Lectures 3-4, we will learn three different PG expressions. Let's try to match the concept of PG with the actual Python implementation. You can refer to the slides **pp. 6-15** of Lecture 4 as a starting point when answering the questions below.

1) PG expression: The **PG.py** is a Python file generated by ChatGPT. This piece of code implements one of the PG expressions. Take a read through the code. Can you figure out which PG expression does ChatGPT implement in **PG.py** (*i.e.*, either (P1), (P2), or (P3))?

2) A deeper look at the code:

- (a) What kind of policy parameterization is ChatGPT using here?
- (b) In Line 35, why does ChatGPT use the “reversed” operation?
- (c) What is Line 41 for?

► Expressions of Policy Gradient (aka Policy Gradient Theorem):

(P1) Total reward: $\nabla_{\theta} V^{\pi_{\theta}}(\mu) = \mathbb{E}_{\tau \sim P_{\mu}^{\pi_{\theta}}} \left[G(\tau) \sum_{t=0}^{\infty} \nabla_{\theta} \log \pi_{\theta}(a_t | s_t) \right]$

(P2) REINFORCE: $\nabla_{\theta} V^{\pi_{\theta}}(\mu) = \mathbb{E}_{\tau \sim P_{\mu}^{\pi_{\theta}}} \left[\sum_{t=0}^{\infty} \gamma^t Q^{\pi_{\theta}}(s_t, a_t) \nabla_{\theta} \log \pi_{\theta}(a_t | s_t) \right]$

(P3) Q-value and discounted state visitation:

$$\nabla_{\theta} V^{\pi_{\theta}}(\mu) = \frac{1}{1-\gamma} \mathbb{E}_{s \sim d_{\mu}^{\pi_{\theta}}} \mathbb{E}_{a \sim \pi_{\theta}(\cdot | s)} \left[Q^{\pi_{\theta}}(s, a) \nabla_{\theta} \log \pi_{\theta}(a | s) \right]$$

3) Do you fully agree with ChatGPT?: Can you either identify 1 bug in the code or provide at least 1 suggestion for improving the code?

Submission Guidelines and Remarks

- Your deliverable shall be a PDF file. Please put all your write-ups into one PDF file (photos/scanned copies are acceptable) and submit your deliverable via E3.
- Please submit it by 9pm, 3/10 (Monday).
- If you have any questions, feel free to post them on Slack.

